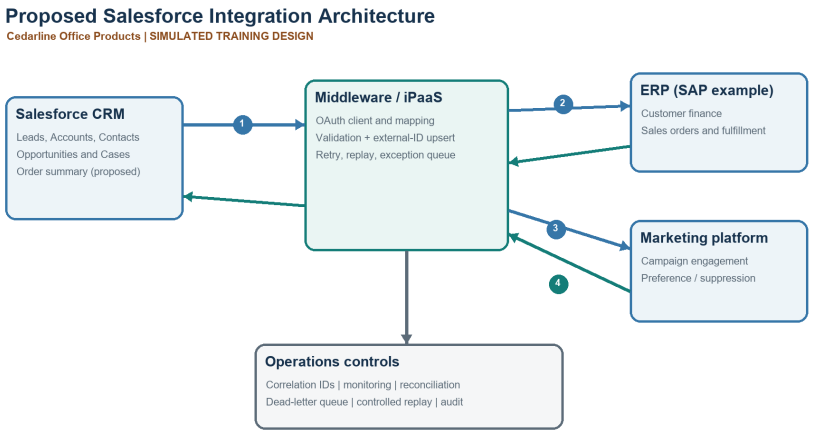
Salesforce Integration Technical Guide

Week 3 quick reference | simulated training scenario

Target architecture



Flows

1 Order: Salesforce Closed Won -> middleware -> ERP; durable acknowledgement returns. Use an asynchronous request/reply pattern, stable external ID, and idempotency key.

2 Status: ERP order/shipment update -> middleware -> CRM summary; match by ERP order ID and reject stale timestamps.

3 Marketing: approved audience -> marketing; engagement and preference changes -> CRM. Apply suppression before activation.

Build and operate checklist

Identity: immutable cross-system IDs and external-ID upsert; do not match on display name alone.

Security: OAuth, dedicated integration identity, least privilege, protected secrets, encrypted transport, redacted logs, data minimization.

API fit: REST for routine transactions; Bulk API 2.0 for large asynchronous jobs; approved event interface for notifications. Confirm org limits and edition.

Reliability: correlation IDs, idempotency, bounded retries/backoff, dead-letter queue, controlled replay, reconciliation.

Governance: system of record, consent precedence, schema version, retention, change approval, and named support owner.

Acceptance checks

- Replay cannot create a duplicate order.
- Invalid IDs and values fail visibly and reach an owner.
- Timeout recovery is traceable and safe to replay.
- Older status cannot overwrite newer status.
- Opt-out is suppressed before audience activation.
- Monitor latency, errors, retries, queue age, freshness, API use, and reconciliation variance.

Proposed objectives to approve

Order acknowledgement <=5 min p95 | Order status <=15 min p95 | Preference suppression <=1 min if supported. Planning goals only; not measured or guaranteed.

Official references

Salesforce Architects Integration Patterns; OAuth / External Client Apps; Bulk API 2.0; Platform Events. Links are listed in the full report.

Fictional scenario; no live system work is represented.